

Exchange Rates for Finite Map Signatures

A Simplified Thermodynamic Complexity

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Abstract

To a surjection between finite sets we assign the multiset of its fiber cardinalities. We compare these signatures by allowing injective preprocessing and bijective relabelling of the outputs, and define an asymptotic exchange rate from Cartesian powers. The rate is expressed by an infimum of ratios of power sums. After the substitution $\epsilon_i = -\log n_i$, the same formula describes homothetic containment of Gibbs energy–entropy regions. The two directed rates do not induce a transitive comparison: $\{5, 3\}$, $\{3, 1, 1\}$, and $\{6, 1\}$ form a strict cycle. Algebraically, this is an ordered semiring resource theory for finite deterministic tasks, related to strong Weihrauch reducibility and to asymptotic tensor restriction.

1 Signatures and implementation

Let $f : X \rightarrow Y$ be onto and enumerate its output as $Y = \{y_1, \dots, y_r\}$. The *fiber* over y_i is the inverse image

$$f^{-1}(y_i) = \{x \in X : f(x) = y_i\}, \quad n_i = |f^{-1}(y_i)|.$$

Order the fiber sizes decreasingly and define the *signature* of f by

$$\sigma(f) = (n_1, \dots, n_r), \quad n_1 \geq \dots \geq n_r \geq 1, \quad r = |Y|.$$

No other information about the map will be used, and we henceforth use the same letter for a map and its signature.

Definition 1 (Implemented by). Let $a = (a_1, \dots, a_r)$ and $b = (b_1, \dots, b_s)$ be decreasing signatures. We say that a is *implemented by* b , and write $a \preceq_{\text{impl}} b$, when $r \leq s$ and

$$a_i \leq b_i \quad (1 \leq i \leq r).$$

For maps, this means that there is an injective input relabeling h_{in} and a bijection from the used output fibers to the output of f such that

$$f = h_{\text{out}} \circ g \circ h_{\text{in}}.$$

More precisely, put

$$X'_g = h_{\text{in}}(X_f), \quad Y'_g = g(X'_g).$$

The two inclusions on the right identify the parts of X_g and Y_g actually used by the implementation, and the following diagram commutes:

$$\begin{array}{ccccc}
X_f & \xrightarrow[\cong]{h_{\text{in}}} & X'_g & \hookrightarrow & X_g \\
f \downarrow & & \downarrow g|_{X'_g} & & \downarrow g \\
Y_f & \xleftarrow[\cong]{h_{\text{out}}} & Y'_g & \hookrightarrow & Y_g.
\end{array}$$

Unused fibers of g may be discarded. Such an implementation exists precisely when each fiber of f injects into its assigned fiber of g , which is the componentwise condition in Definition 1.

Cartesian product multiplies fiber sizes:

$$a \otimes b = \text{sort}\{a_i b_j : 1 \leq i \leq r, 1 \leq j \leq s\}.$$

Multiplicity is retained, and $a^{\otimes k}$ denotes the k -fold product.

Disjoint union supplies a second operation:

$$f \sqcup g : X_f \sqcup X_g \longrightarrow Y_f \sqcup Y_g$$

has signature

$$a + b = \text{sort}(a_1, \dots, a_r, b_1, \dots, b_s).$$

Both operations respect implementation, and Cartesian product distributes over this sum:

$$a \preceq_{\text{impl}} b, c \preceq_{\text{impl}} d \implies a + c \preceq_{\text{impl}} b + d, \quad a \otimes c \preceq_{\text{impl}} b \otimes d, \quad a \otimes (b + c) = (a \otimes b) + (a \otimes c).$$

After adjoining the empty signature as zero, signatures form an ordered commutative semiring with multiplicative unit $\{1\}$. Here $+$ is a classically labelled alternative and $\otimes$ is parallel composition. Only the latter enters the exchange rate.

Definition 2 (Exchange rate). For signatures f and g , set

$$k_{\max}^{g \leftarrow f}(n) = \max \left\{ k \geq 0 : f^{\otimes k} \preceq_{\text{impl}} g^{\otimes n} \right\}.$$

The exchange rate, with the implementer written first, is

$$C(g \mid f) = \lim_{n \rightarrow \infty} \frac{k_{\max}^{g \leftarrow f}(n)}{n}.$$

Tensoring optimal implementations at block lengths n and m gives

$$k_{\max}(n + m) \geq k_{\max}(n) + k_{\max}(m).$$

Fekete's lemma now yields

$$\lim_{n \rightarrow \infty} \frac{k_{\max}(n)}{n} = \sup_{n \geq 1} \frac{k_{\max}(n)}{n},$$

with $+\infty$ allowed. The limit is the operational definition of $C(g \mid f)$.

2 Related frameworks

The operations above place the construction among semiring resource theories: implementation is the free conversion, $+$ is disjoint choice, and $\otimes$ combines independent resources. General accounts of resource theories and their ordered algebraic structures are given in [1, 2].

The one-shot factorization

$$f = h_{\text{out}} \circ g \circ h_{\text{in}}$$

has the form of strong Weihrauch reduction: one call to g , preceded and followed by fixed processors [3]. Unlike the computable processors in Weihrauch theory, ours are injective or bijective on the sets actually used. This restriction preserves the fiber capacities.

The asymptotic construction is closer to tensor restriction. Tensors encode bilinear tasks, linear maps implement restrictions, and tensor product combines copies. Strassen’s asymptotic spectrum characterizes large tensor powers by multiplicative monotones [4]. The relative exponent

$$\omega(t, s) = \lim_{n \rightarrow \infty} \frac{1}{n} \min\{m : t^{\otimes m} \text{ restricts to } s^{\otimes n}\}$$

measures resource cost per target copy; it is used, for example, to quantify irreversibility in matrix multiplication [5]. Its convention is reciprocal to ours after reversing the conversion. Here the spectral monotones are the power sums $Z_a(\beta)$. The resulting formula is also analogous to exact asymptotic majorization, where the rate is a minimum of Rényi-entropy ratios [6].

Channel simulation provides a third comparison: an encoder and decoder use copies of one channel to reproduce another. With common-randomness assistance, the first-order rate for point-to-point discrete memoryless channels is the ratio of their capacities [7]. Appendix A recovers the corresponding logarithmic cardinality ratio when both processors in the finite-map model are left unrestricted.

3 Power sums and Gibbs curves

For a signature $a = (a_i)$, define its power-sum partition function

$$Z_a(\beta) = \sum_i a_i^\beta, \quad \beta \geq 0,$$

where $\beta = 1/T$ is inverse temperature. The semiring operations become ordinary addition and multiplication:

$$Z_{a+b}(\beta) = Z_a(\beta) + Z_b(\beta), \quad Z_{a \otimes b}(\beta) = Z_a(\beta) Z_b(\beta).$$

The fibers of the k -fold Cartesian power $a^{\otimes k}$ are indexed by $(i_1, \dots, i_k)$ and have sizes $a_{i_1} \cdots a_{i_k}$. Therefore

$$\begin{aligned} Z_{a^{\otimes k}}(\beta) &= \sum_{i_1, \dots, i_k} (a_{i_1} \cdots a_{i_k})^\beta \\ &= \left(\sum_i a_i^\beta \right)^k = Z_a(\beta)^k, \quad \log Z_{a^{\otimes k}}(\beta) = k \log Z_a(\beta). \end{aligned} \tag{1}$$

Proposition 1. Let f and g each have at least two fibers and a fiber of size greater than one. Then

$$C(g \mid f) = \inf_{\beta \in [0, \infty]} \frac{\log Z_g(\beta)}{\log Z_f(\beta)}, \tag{2}$$

where the value at $\beta = \infty$ is

$$\frac{\log \max_j g_j}{\log \max_i f_i}.$$

Proof sketch. If $f^{\otimes k} \preceq_{\text{impl}} g^{\otimes n}$, applying the power sum at every $\beta \geq 0$ gives

$$Z_f(\beta)^k \leq Z_g(\beta)^n, \quad \frac{k}{n} \leq \frac{\log Z_g(\beta)}{\log Z_f(\beta)}.$$

The endpoint $\beta = \infty$ gives the same inequality for the largest fibers, so the right-hand side of (2) is an upper bound.

For the converse, choose a rational k/n strictly below that infimum. Sort the logarithms of the fibers in $f^{\otimes k\ell}$ and $g^{\otimes n\ell}$. The exponential growth rate of the number of entries above a fixed threshold is the Legendre transform of $\log Z_f$, respectively $\log Z_g$. The strict power-sum inequalities, including the two endpoints, therefore give componentwise domination for all sufficiently large ℓ . Hence $f^{\otimes k\ell} \preceq_{\text{impl}} g^{\otimes n\ell}$. Letting k/n increase to the infimum proves the reverse inequality. This is the type-counting argument used in exact asymptotic majorization; compare [6]. $\square$

Put

$$\epsilon_i = -\log a_i, \quad p_i(T) = \frac{e^{-\epsilon_i/T}}{\sum_j e^{-\epsilon_j/T}} = \frac{a_i^{1/T}}{Z_a(1/T)}.$$

The corresponding mean energy and entropy are

$$E_a(T) = \sum_i p_i(T) \epsilon_i, \quad H_a(T) = -\sum_i p_i(T) \log p_i(T).$$

They satisfy

$$\log Z_a(1/T) = H_a(T) - \frac{E_a(T)}{T}.$$

Since the Gibbs distribution of a Cartesian power is a product distribution,

$$H_{a^{\otimes k}}(T) = kH_a(T), \quad E_{a^{\otimes k}}(T) = kE_a(T),$$

and therefore

$$E_{a^{\otimes k}}(H) = kE_a(H/k),$$

so Cartesian powers act by homothety in the (E, H) -plane.

Let Γ_a denote the Gibbs curve $\{(E_a(T), H_a(T)) : 0 < T < \infty\}$, completed by its endpoint segments, and let $\mathcal{R}_a$ be the closed convex region between this boundary and the line $E = 0$. The quantity

$$\log Z_a(\beta) = H_a(T) - \beta E_a(T)$$

is the offset of the supporting line with normal $(-\beta, 1)$. Proposition 1 is therefore equivalent to

$$C(g \mid f) = \sup\{c \geq 0 : c\mathcal{R}_f \subseteq \mathcal{R}_g\}.$$

Equivalently, $C(g \mid f)^{-1}\Gamma_g$ is the smallest homothetic copy of Γ_g whose region contains Γ_f .

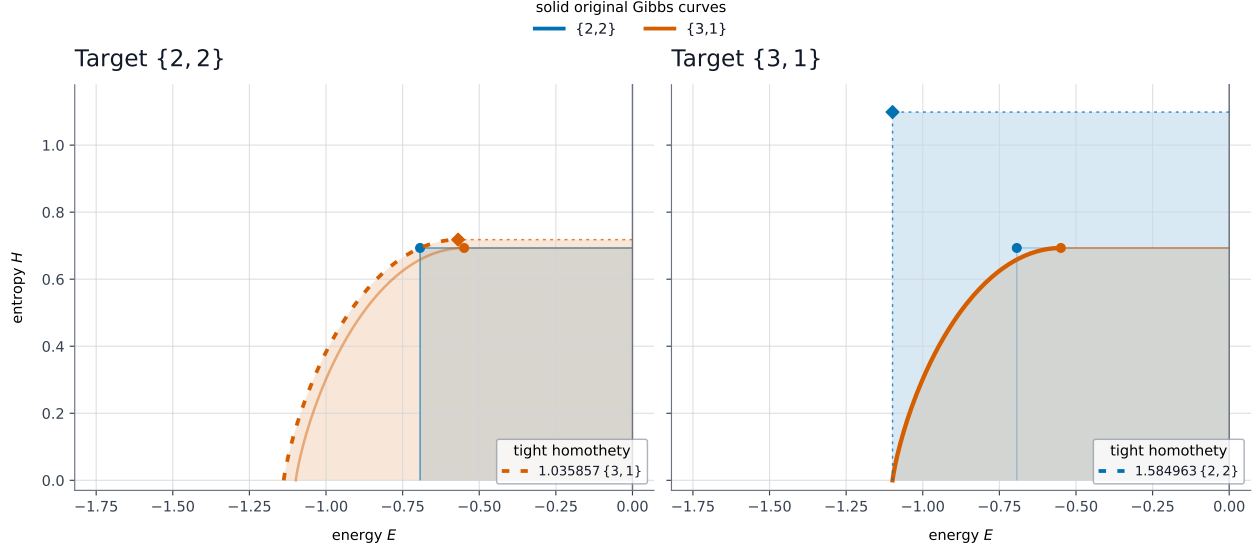


Figure 1: Exchange rates as planar dilations. In each panel, one energy–entropy region is scaled about the origin until it just contains the other. Thin segments at the $T = 0$ and $T = \infty$ endpoints close the regions; dotted boundaries denote the scaled copy, and translucent fills make the containment visible. The required scale factors are $1/C(\{3, 1\} \mid \{2, 2\}) = 1.03585676\dots$ on the left and $1/C(\{2, 2\} \mid \{3, 1\}) = \log 3 / \log 2$ on the right.

4 The pair $\{2, 2\}$ and $\{3, 1\}$

Let $g = \{2, 2\}$ and $f = \{3, 1\}$. Every fiber of $g^{\otimes n}$ has size 2^n . The largest fiber of $f^{\otimes k}$ has size 3^k , and the remaining rank constraint is weaker at the optimum. Hence

$$C(\{2, 2\} \mid \{3, 1\}) = \frac{\log 2}{\log 3} = 0.6309297536\dots$$

The reverse direction is limited at a finite temperature:

$$\begin{aligned} C(\{3, 1\} \mid \{2, 2\}) &= \min_{\beta \geq 0} \frac{\log(3^\beta + 1)}{(\beta + 1) \log 2} \\ &= 0.9653844417\dots, \quad \beta_* = 0.4036799030\dots, \quad T_* = \beta_*^{-1} = 2.47721027\dots \end{aligned}$$

Figure 1 displays both homotheties. Solid curves are unscaled; dotted curves are enlarged by the reciprocal directed rate. Thin endpoint segments complete the boundaries, and the fills show the regions $\mathcal{R}_a$. Because the two levels of $\{2, 2\}$ coincide, its Gibbs curve reduces to the point $(-\log 2, \log 2)$.

5 A two-sided finite-temperature tangency

Both directed rates may be attained away from the endpoints. Consider

$$a = \{6, 5, 2, 1\}, \quad b = \{6, 4, 3, 2\}.$$

The two signatures have equal length and equal largest fiber, so both endpoint ratios are one:

$$\frac{\log Z_a(0)}{\log Z_b(0)} = 1, \quad \lim_{\beta \rightarrow \infty} \frac{\log Z_a(\beta)}{\log Z_b(\beta)} = \frac{\log 6}{\log 6} = 1.$$

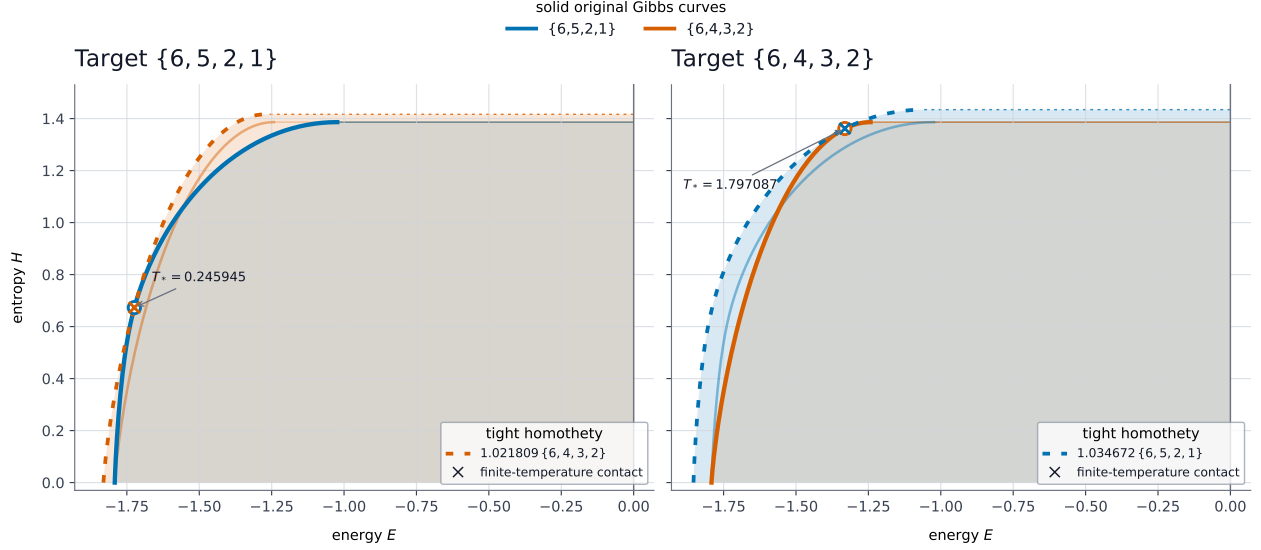


Figure 2: Two-sided finite-temperature tangency. The left panel scales b by $1/C(b \mid a)$ to embrace a ; the right panel scales a by $1/C(a \mid b)$ to embrace b . Crosses mark the interior contact points. Thin dotted horizontal lines close the scaled regions at their $T = \infty$ endpoints; translucent fills extend to $E = 0$.

The interior minima are:

Direction	C	T_*	$\beta_* = 1/T_*$
$C(a \mid b)$	0.9664896992	1.7970865612	0.5564562229
$C(b \mid a)$	0.9786567233	0.2459449336	4.0659508025

If

$$R(\beta) = \frac{\log Z_g(\beta)}{\log Z_f(\beta)}$$

has an interior minimum at β_* , then $R'(\beta_*) = 0$ and $\log Z = H - \beta E$ give

$$E_g(T_*) = C(g \mid f)E_f(T_*), \quad H_g(T_*) = C(g \mid f)H_f(T_*).$$

The target curve and the scaled source curve therefore meet at the same point. Their slopes also agree, since

$$\frac{dH}{dE} = \frac{1}{T_*} = \beta_*,$$

and this slope is invariant under homothety. The two contacts in Figure 2 are genuine tangencies.

6 A cyclic strict comparison

Definition 3. For signatures a, b , define

$$a \prec b \iff C(a \mid b) < C(b \mid a).$$

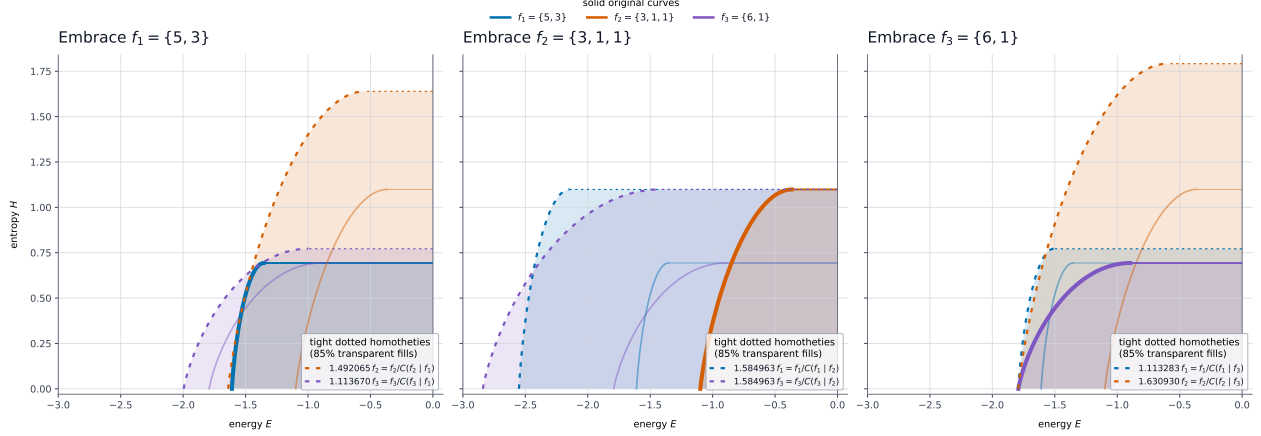


Figure 3: The cyclic example. Solid and dotted boundaries are original and scaled, respectively. Horizontal $T = \infty$ closures and 85%-transparent fills extending to $E = 0$ display target containment.

In this terminology, b is more complex than a when conversion from b to a is more efficient than conversion in the opposite direction. The relation is not transitive.

Set

$$f_1 = \{5, 3\}, \quad f_2 = \{3, 1, 1\}, \quad f_3 = \{6, 1\}.$$

Equation (2) gives the comparisons in Table 1.

Table 1: Directed rates producing a strict cycle.

Pair	Forward rate	Reverse rate	Conclusion
f_1, f_2	$C(f_1 f_2) = 0.630929754$	$C(f_2 f_1) = 0.670211928$	$f_1 \prec f_2$
f_2, f_3	$C(f_2 f_3) = 0.613147193$	$C(f_3 f_2) = 0.630929754$	$f_2 \prec f_3$
f_3, f_1	$C(f_3 f_1) = 0.897931851$	$C(f_1 f_3) = 0.898244402$	$f_3 \prec f_1$

The three rows give

$$f_1 \prec f_2, \quad f_2 \prec f_3, \quad f_3 \prec f_1.$$

Hence $\prec$ is not a strict order. Figure 3 shows the three comparisons geometrically. In the panel with target f_i , the dotted curve for f_j is $C(f_j | f_i)^{-1} \Gamma_{f_j}$.

7 A thermodynamic analogy and its limits

Associate signatures $A = (a_i)$ and $B = (b_j)$ with two idealized particle species whose internal energy levels are

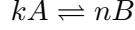
$$\epsilon_i^A = -\log a_i, \quad \epsilon_j^B = -\log b_j.$$

Then $A^{\otimes k}$ and $B^{\otimes n}$ represent k noninteracting particles of type A and n noninteracting particles of type B , respectively. If their scaled Gibbs curves are tangent at interior points,

$$\frac{dH_A}{dE_A} = \frac{dH_B}{dE_B} = \frac{1}{T}.$$

The tangent Gibbs states therefore have the same temperature.

This identification is kinematic, not dynamical: $\preccurlyeq_{\text{impl}}$ records an allowed combinatorial conversion, not a reaction mechanism. For the formal stoichiometry



write $r = k/n$. In the asymptotic closure,

$$B^{\otimes n} \longrightarrow A^{\otimes k} \iff r \leq C(B \mid A), \quad (3)$$

$$A^{\otimes k} \longrightarrow B^{\otimes n} \iff r \geq \frac{1}{C(A \mid B)}. \quad (4)$$

Strict inequalities guarantee the corresponding direction for sufficiently large block sizes; an exact boundary ratio need not be attained at finite size. Since

$$C(A \mid B)C(B \mid A) \leq 1,$$

the thresholds leave the interval

$$C(B \mid A) \leq r \leq \frac{1}{C(A \mid B)}.$$

For ratios in its interior, neither direction need realize the prescribed stoichiometry. Coincident endpoints mean asymptotic reversibility; otherwise the interval measures irreversibility.

Physical equilibrium requires more. At fixed temperature and pressure the reaction affinity must vanish:

$$\mathcal{A} = k\mu_A - n\mu_B = 0,$$

where μ_A and μ_B are chemical potentials. Neither chemical potentials nor reaction kinetics occur in our definition. Equations (3)–(4) therefore describe allowed conversions, not spontaneous reaction directions.

The strict cycle admits a formal comparison with a driven reaction network. For $a \xrightarrow{s} b$, set

$$\alpha(a \rightarrow b) = \log \frac{C(a \mid b)}{C(b \mid a)} > 0.$$

For the cycle of Section 6,

$$f_1 \xrightarrow{s} f_3 \xrightarrow{s} f_2 \xrightarrow{s} f_1,$$

the sum around the cycle is

$$\begin{aligned} \mathcal{A}_{\text{cyc}} &= \alpha(f_1 \rightarrow f_3) + \alpha(f_3 \rightarrow f_2) + \alpha(f_2 \rightarrow f_1) \\ &= \log \frac{0.898244402}{0.897931851} + \log \frac{0.630929754}{0.613147193} + \log \frac{0.670211928}{0.630929754} \\ &= 0.089337, \quad e^{\mathcal{A}_{\text{cyc}}} = 1.09345. \end{aligned}$$

If α is read as a logarithmic forward–reverse bias, the orientation matches the reaction network

$$A \rightleftharpoons B, \quad B \rightleftharpoons C, \quad C \rightleftharpoons A.$$

It cannot represent circulation in a closed equilibrium gas. Reaction affinity is the negative derivative of Gibbs energy with respect to reaction extent [8]; detailed balance forces the affinity around a closed cycle to vanish. Nonzero cycle affinity instead belongs to an open system maintained

by reservoirs. Such networks are studied in [9, 10]; a stochastic three-species example appears in [11].

The analogy stops at this structural resemblance. The rates $C(a | b)$ are not kinetic constants, and the three minima may occur at different temperatures. Without kinetics, conservation laws, or external work accounting, the strict cycle predicts neither a current nor an oscillation.

A fixed reaction energy can be included as an offset. For

$$kA \rightleftharpoons nB + q$$

shift every level of B by q/n :

$$\tilde{\epsilon}_{B,j} = \epsilon_{B,j} + \frac{q}{n}.$$

Then

$$\tilde{E}_B(H) = E_B(H) + \frac{q}{n}, \quad \tilde{Z}_B(1/T) = e^{-q/(nT)} Z_B(1/T),$$

while $H_B(T)$ and $dH_B/dE = 1/T$ are unchanged. The n -particle region is therefore translated horizontally by q . This accounts for a fixed offset carried by a work-storage degree of freedom; heat transfer to a bath cannot be represented by a horizontal shift alone.

The model omits interactions, quantum statistics, translational and volume degrees of freedom, mixing entropy, conservation laws, kinetics, activation barriers, and reservoirs. Its energies are dimensionless ($k_B = 1$), and relative offsets between particle types must be fixed whenever particle number can change. These omissions delimit the analogy.

The supporting function

$$\Psi_A(T) = \log Z_A(1/T) = -\frac{F_A(T)}{T}$$

is the Massieu potential. Proposition 1 becomes

$$C(A | B) = \inf_{T>0} \frac{\Psi_A(T)}{\Psi_B(T)}$$

and asks for the largest c such that $\Psi_A(T) \geq c\Psi_B(T)$ for every T . An interior minimizer is the bottleneck temperature. Sparaciari, Oppenheim, and Fritz study a related resource theory without a fixed background temperature, in which asymptotic rates are read from an energy–entropy diagram [12]. Our comparison uses the whole Gibbs boundary rather than a single prescribed heat bath.

An exact physical homothety occurs for ultrarelativistic equilibrium sectors. If g_X is the effective number of internal degrees of freedom, then in natural units and at fixed volume

$$E_X(T) = \frac{\pi^2}{30} g_X V T^4, \quad H_X(T) = \frac{2\pi^2}{45} g_X V T^3.$$

Therefore

$$E_A(T) = \frac{g_A}{g_B} E_B(T), \quad H_A(T) = \frac{g_A}{g_B} H_B(T)$$

at every temperature, so the two boundaries are homothetic with factor g_A/g_B . For photons $g_\gamma = 2$; for an ultrarelativistic electron–positron sector $g_{e^\pm} = (7/8) \times 4 = 7/2$. The factor is therefore $7/4$, as in Figure 4. In the early universe, $e^- + e^+ \rightleftharpoons 2\gamma$ maintained equilibrium until the temperature fell below the electron mass and annihilation transferred entropy to the photon bath [13].

For general Gibbs systems, homothetic containment is at present a proposed complexity measure. Infinite spectra and an operational conversion model require separate treatment.

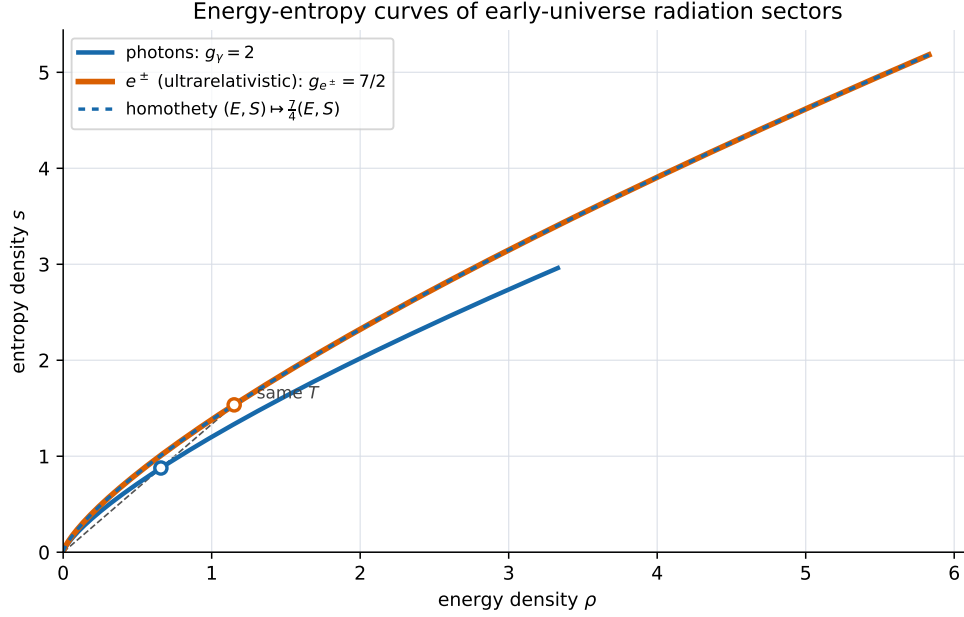


Figure 4: Exact homothety in the ultrarelativistic limit. At fixed temperature, the electron–positron sector has $7/4$ times the energy and entropy densities of the photon sector; the rescaled photon curve therefore coincides with it.

8 Conclusion

A fiber signature is a coarse invariant of a finite map, but Cartesian powers already give it a nontrivial asymptotic conversion theory. The operational rate, the power-sum formula, and containment of Gibbs regions are equivalent descriptions of that theory. The directed rates cannot be compressed into a global ranking, as the explicit three-cycle shows. Extending Gibbs-region containment beyond finite signatures remains plausible, but it requires an operational model distinct from chemical reaction dynamics.

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A Unrestricted input and output maps

Suppose that the maps h_{in} and h_{out} in the definition of implementation are unrestricted. Then b is implementable by a if and only if

$$|Y_b| \leq |Y_a|.$$

An arbitrary h_{in} may collapse each fiber of b onto one chosen input of a , so only the number of available output fibers remains. A deterministic h_{out} cannot increase that number.

For Cartesian powers the condition is therefore

$$|Y_b|^k \leq |Y_a|^n, \quad k_{\max}(n) = \left\lfloor n \frac{\log |Y_a|}{\log |Y_b|} \right\rfloor.$$

Hence, for $|Y_a|, |Y_b| > 1$,

$$C(a | b) = \frac{\log |Y_a|}{\log |Y_b|}.$$

The same ratio appears in assisted point-to-point channel interconversion [7]: a surjective deterministic channel with output set Y has capacity $\log |Y|$. Unrestricted processors retain only this capacity; the processors of Definition 1 also retain the individual fiber sizes.

$C(a | b)C(b | a) = 1$, so the comparison from Definition 3 reduces to

$$a \prec b \iff |Y_a| < |Y_b|.$$

This is a total preorder, and quotienting by equal output cardinality gives a linear order. In partition-function language only $Z_a(0) = |Y_a|$ remains. If $|Y_b| = 1$, then $C(a \mid b) = +\infty$; if $|Y_a| = 1 < |Y_b|$, then $C(a \mid b) = 0$.

B The first 69 non-special signatures

We omit the exceptional infinite families $\{N\}$ and $\{1, \dots, 1\}$. To enumerate every other signature without imposing a fixed length or entry bound, put

$$\mathcal{S}_B = \{a : |a| \geq 2, a_1 > 1, a_1 + 2|a| \leq B\}.$$

Every non-special signature eventually belongs to $\mathcal{S}_B$. For each B , the strongly connected components of the strict comparison graph are contracted and topologically ordered by the deterministic rule implemented in the analysis code. At the stated screening precision, the first 69 entries agree for $B = 18, 19, 20$. Table 2 records this stable prefix; its rates were recomputed from the high-accuracy cache.

Because $\prec$ is not transitive, this list is not a linear extension of all pairwise comparisons. A row for x in Table 3 records each earlier a for which

$$C(x \mid a) < C(a \mid x)$$

despite the displayed order $a < x$.

Table 2: The stabilized first 69 non-special signatures in the deterministic condensation-DAG order.

n	signature	n	signature	n	signature
1	{2, 1}	24	{6, 6}	47	{9, 8}
2	{2, 2}	25	{7, 1}	48	{9, 9}
3	{3, 1}	26	{7, 2}	49	{10, 1}
4	{2, 1, 1}	27	{7, 3}	50	{10, 2}
5	{3, 2}	28	{7, 4}	51	{10, 3}
6	{2, 2, 1}	29	{7, 5}	52	{10, 4}
7	{3, 3}	30	{7, 6}	53	{10, 5}
8	{2, 1, 1, 1}	31	{7, 7}	54	{10, 6}
9	{2, 2, 2}	32	{8, 1}	55	{10, 7}
10	{4, 1}	33	{8, 2}	56	{10, 8}
11	{4, 2}	34	{8, 3}	57	{10, 9}
12	{4, 3}	35	{8, 4}	58	{10, 10}
13	{4, 4}	36	{8, 5}	59	{11, 1}
14	{5, 1}	37	{8, 6}	60	{11, 2}
15	{5, 2}	38	{8, 7}	61	{11, 3}
16	{5, 3}	39	{8, 8}	62	{11, 4}
17	{5, 4}	40	{9, 1}	63	{11, 5}
18	{5, 5}	41	{9, 2}	64	{11, 6}
19	{6, 1}	42	{9, 3}	65	{11, 7}
20	{6, 2}	43	{9, 4}	66	{11, 8}
21	{6, 3}	44	{9, 5}	67	{11, 9}
22	{6, 4}	45	{9, 6}	68	{11, 10}
23	{6, 5}	46	{9, 7}	69	{11, 11}

Table 3: Backward comparisons that obstruct a linear order.

n	x	exceptions with previous n_a values
14	$\{5, 1\}$	$n_a = 13, a = \{4, 4\}$: $C(x a) = 0.844794, C(a x) = 0.861353$
19	$\{6, 1\}$	$n_a = 16, a = \{5, 3\}$: $C(x a) = 0.897932, C(a x) = 0.898244$ $n_a = 17, a = \{5, 4\}$: $C(x a) = 0.857550, C(a x) = 0.898244$ $n_a = 18, a = \{5, 5\}$: $C(x a) = 0.825066, C(a x) = 0.898244$
25	$\{7, 1\}$	$n_a = 21, a = \{6, 3\}$: $C(x a) = 0.902627, C(a x) = 0.920782$ $n_a = 22, a = \{6, 4\}$: $C(x a) = 0.866396, C(a x) = 0.920782$ $n_a = 23, a = \{6, 5\}$: $C(x a) = 0.837432, C(a x) = 0.920782$ $n_a = 24, a = \{6, 6\}$: $C(x a) = 0.813202, C(a x) = 0.920782$
26	$\{7, 2\}$	$n_a = 23, a = \{6, 5\}$: $C(x a) = 0.906054, C(a x) = 0.920782$ $n_a = 24, a = \{6, 6\}$: $C(x a) = 0.878094, C(a x) = 0.920782$
32	$\{8, 1\}$	$n_a = 24, a = \{6, 6\}$: $C(x a) = 0.844632, C(a x) = 0.861654$ $n_a = 27, a = \{7, 3\}$: $C(x a) = 0.906390, C(a x) = 0.935785$ $n_a = 28, a = \{7, 4\}$: $C(x a) = 0.873076, C(a x) = 0.935785$ $n_a = 29, a = \{7, 5\}$: $C(x a) = 0.846552, C(a x) = 0.935785$ $n_a = 30, a = \{7, 6\}$: $C(x a) = 0.824430, C(a x) = 0.935785$ $n_a = 31, a = \{7, 7\}$: $C(x a) = 0.805397, C(a x) = 0.935785$
33	$\{8, 2\}$	$n_a = 29, a = \{7, 5\}$: $C(x a) = 0.911301, C(a x) = 0.935785$ $n_a = 30, a = \{7, 6\}$: $C(x a) = 0.886261, C(a x) = 0.935785$ $n_a = 31, a = \{7, 7\}$: $C(x a) = 0.864389, C(a x) = 0.935785$

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Table 3 continued from the previous page

n	x	exceptions with previous n_a values
34	$\{8, 3\}$	$n_a = 30, a = \{7, 6\}$: $C(x a) = 0.930247, C(a x) = 0.935785$ $n_a = 31, a = \{7, 7\}$: $C(x a) = 0.906390, C(a x) = 0.935785$
40	$\{9, 1\}$	$n_a = 29, a = \{7, 5\}$: $C(x a) = 0.871798, C(a x) = 0.885622$ $n_a = 30, a = \{7, 6\}$: $C(x a) = 0.850314, C(a x) = 0.885622$ $n_a = 31, a = \{7, 7\}$: $C(x a) = 0.831776, C(a x) = 0.885622$ $n_a = 34, a = \{8, 3\}$: $C(x a) = 0.909518, C(a x) = 0.946395$ $n_a = 35, a = \{8, 4\}$: $C(x a) = 0.878388, C(a x) = 0.946395$ $n_a = 36, a = \{8, 5\}$: $C(x a) = 0.853673, C(a x) = 0.946395$ $n_a = 37, a = \{8, 6\}$: $C(x a) = 0.833105, C(a x) = 0.946395$ $n_a = 38, a = \{8, 7\}$: $C(x a) = 0.815439, C(a x) = 0.946395$ $n_a = 39, a = \{8, 8\}$: $C(x a) = 0.799923, C(a x) = 0.946395$
41	$\{9, 2\}$	$n_a = 35, a = \{8, 4\}$: $C(x a) = 0.942478, C(a x) = 0.946395$ $n_a = 36, a = \{8, 5\}$: $C(x a) = 0.915285, C(a x) = 0.946395$ $n_a = 37, a = \{8, 6\}$: $C(x a) = 0.892295, C(a x) = 0.946395$ $n_a = 38, a = \{8, 7\}$: $C(x a) = 0.872314, C(a x) = 0.946395$ $n_a = 39, a = \{8, 8\}$: $C(x a) = 0.854578, C(a x) = 0.946395$
42	$\{9, 3\}$	$n_a = 37, a = \{8, 6\}$: $C(x a) = 0.933809, C(a x) = 0.946395$ $n_a = 38, a = \{8, 7\}$: $C(x a) = 0.912294, C(a x) = 0.946395$ $n_a = 39, a = \{8, 8\}$: $C(x a) = 0.892960, C(a x) = 0.946395$
43	$\{9, 4\}$	$n_a = 38, a = \{8, 7\}$: $C(x a) = 0.944850, C(a x) = 0.946395$ $n_a = 39, a = \{8, 8\}$: $C(x a) = 0.924272, C(a x) = 0.946395$

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Table 3 continued from the previous page

n	x	exceptions with previous n_a values
49	$\{10, 1\}$	$n_a = 35, a = \{8, 4\}$: $C(x a) = 0.899143, C(a x) = 0.903090$ $n_a = 36, a = \{8, 5\}$: $C(x a) = 0.875102, C(a x) = 0.903090$ $n_a = 37, a = \{8, 6\}$: $C(x a) = 0.855010, C(a x) = 0.903090$ $n_a = 38, a = \{8, 7\}$: $C(x a) = 0.837714, C(a x) = 0.903090$ $n_a = 39, a = \{8, 8\}$: $C(x a) = 0.822502, C(a x) = 0.903090$ $n_a = 41, a = \{9, 2\}$: $C(x a) = 0.952333, C(a x) = 0.954243$ $n_a = 42, a = \{9, 3\}$: $C(x a) = 0.912182, C(a x) = 0.954243$ $n_a = 43, a = \{9, 4\}$: $C(x a) = 0.882762, C(a x) = 0.954243$ $n_a = 44, a = \{9, 5\}$: $C(x a) = 0.859454, C(a x) = 0.954243$ $n_a = 45, a = \{9, 6\}$: $C(x a) = 0.840087, C(a x) = 0.954243$ $n_a = 46, a = \{9, 7\}$: $C(x a) = 0.823475, C(a x) = 0.954243$ $n_a = 47, a = \{9, 8\}$: $C(x a) = 0.808903, C(a x) = 0.954243$ $n_a = 48, a = \{9, 9\}$: $C(x a) = 0.795900, C(a x) = 0.954243$
50	$\{10, 2\}$	$n_a = 38, a = \{8, 7\}$: $C(x a) = 0.894640, C(a x) = 0.903090$ $n_a = 39, a = \{8, 8\}$: $C(x a) = 0.877498, C(a x) = 0.903090$ $n_a = 43, a = \{9, 4\}$: $C(x a) = 0.943960, C(a x) = 0.954243$ $n_a = 44, a = \{9, 5\}$: $C(x a) = 0.918479, C(a x) = 0.954243$ $n_a = 45, a = \{9, 6\}$: $C(x a) = 0.897026, C(a x) = 0.954243$ $n_a = 46, a = \{9, 7\}$: $C(x a) = 0.878445, C(a x) = 0.954243$ $n_a = 47, a = \{9, 8\}$: $C(x a) = 0.862002, C(a x) = 0.954243$ $n_a = 48, a = \{9, 9\}$: $C(x a) = 0.847215, C(a x) = 0.954243$

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n	x	exceptions with previous n_a values
51	$\{10, 3\}$	$n_a = 45, a = \{9, 6\}$: $C(x a) = 0.936518, C(a x) = 0.954243$ $n_a = 46, a = \{9, 7\}$: $C(x a) = 0.916673, C(a x) = 0.954243$ $n_a = 47, a = \{9, 8\}$: $C(x a) = 0.898942, C(a x) = 0.954243$ $n_a = 48, a = \{9, 9\}$: $C(x a) = 0.882860, C(a x) = 0.954243$
52	$\{10, 4\}$	$n_a = 46, a = \{9, 7\}$: $C(x a) = 0.947468, C(a x) = 0.954243$ $n_a = 47, a = \{9, 8\}$: $C(x a) = 0.928787, C(a x) = 0.954243$ $n_a = 48, a = \{9, 9\}$: $C(x a) = 0.911656, C(a x) = 0.954243$
53	$\{10, 5\}$	$n_a = 48, a = \{9, 9\}$: $C(x a) = 0.936626, C(a x) = 0.954243$

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Table 3 continued from the previous page

n	x	exceptions with previous n_a values
59	$\{11, 1\}$	$n_a = 38, a = \{8, 7\}$: $C(x a) = 0.856507, C(a x) = 0.867194$ $n_a = 39, a = \{8, 8\}$: $C(x a) = 0.841648, C(a x) = 0.867194$ $n_a = 43, a = \{9, 4\}$: $C(x a) = 0.900788, C(a x) = 0.916314$ $n_a = 44, a = \{9, 5\}$: $C(x a) = 0.878004, C(a x) = 0.916314$ $n_a = 45, a = \{9, 6\}$: $C(x a) = 0.859006, C(a x) = 0.916314$ $n_a = 46, a = \{9, 7\}$: $C(x a) = 0.842681, C(a x) = 0.916314$ $n_a = 47, a = \{9, 8\}$: $C(x a) = 0.828343, C(a x) = 0.916314$ $n_a = 48, a = \{9, 9\}$: $C(x a) = 0.815542, C(a x) = 0.916314$ $n_a = 50, a = \{10, 2\}$: $C(x a) = 0.952868, C(a x) = 0.960253$ $n_a = 51, a = \{10, 3\}$: $C(x a) = 0.914492, C(a x) = 0.960253$ $n_a = 52, a = \{10, 4\}$: $C(x a) = 0.886455, C(a x) = 0.960253$ $n_a = 53, a = \{10, 5\}$: $C(x a) = 0.864279, C(a x) = 0.960253$ $n_a = 54, a = \{10, 6\}$: $C(x a) = 0.845875, C(a x) = 0.960253$ $n_a = 55, a = \{10, 7\}$: $C(x a) = 0.830106, C(a x) = 0.960253$ $n_a = 56, a = \{10, 8\}$: $C(x a) = 0.816285, C(a x) = 0.960253$ $n_a = 57, a = \{10, 9\}$: $C(x a) = 0.803964, C(a x) = 0.960253$ $n_a = 58, a = \{10, 10\}$: $C(x a) = 0.792835, C(a x) = 0.960253$

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n	x	exceptions with previous n_a values
60	$\{11, 2\}$	$n_a = 45, a = \{9, 6\}$: $C(x a) = 0.915642, C(a x) = 0.916314$ $n_a = 46, a = \{9, 7\}$: $C(x a) = 0.897629, C(a x) = 0.916314$ $n_a = 47, a = \{9, 8\}$: $C(x a) = 0.881636, C(a x) = 0.916314$ $n_a = 48, a = \{9, 9\}$: $C(x a) = 0.867225, C(a x) = 0.916314$ $n_a = 52, a = \{10, 4\}$: $C(x a) = 0.945256, C(a x) = 0.960253$ $n_a = 53, a = \{10, 5\}$: $C(x a) = 0.921131, C(a x) = 0.960253$ $n_a = 54, a = \{10, 6\}$: $C(x a) = 0.900885, C(a x) = 0.960253$ $n_a = 55, a = \{10, 7\}$: $C(x a) = 0.883391, C(a x) = 0.960253$ $n_a = 56, a = \{10, 8\}$: $C(x a) = 0.867946, C(a x) = 0.960253$ $n_a = 57, a = \{10, 9\}$: $C(x a) = 0.854085, C(a x) = 0.960253$ $n_a = 58, a = \{10, 10\}$: $C(x a) = 0.841485, C(a x) = 0.960253$
61	$\{11, 3\}$	$n_a = 48, a = \{9, 9\}$: $C(x a) = 0.902904, C(a x) = 0.916314$ $n_a = 53, a = \{10, 5\}$: $C(x a) = 0.959953, C(a x) = 0.960253$ $n_a = 54, a = \{10, 6\}$: $C(x a) = 0.938697, C(a x) = 0.960253$ $n_a = 55, a = \{10, 7\}$: $C(x a) = 0.920119, C(a x) = 0.960253$ $n_a = 56, a = \{10, 8\}$: $C(x a) = 0.903587, C(a x) = 0.960253$ $n_a = 57, a = \{10, 9\}$: $C(x a) = 0.888649, C(a x) = 0.960253$ $n_a = 58, a = \{10, 10\}$: $C(x a) = 0.874982, C(a x) = 0.960253$

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n	x	exceptions with previous n_a values
62	$\{11, 4\}$	$n_a = 55, a = \{10, 7\}$: $C(x a) = 0.949460, C(a x) = 0.960253$ $n_a = 56, a = \{10, 8\}$: $C(x a) = 0.932147, C(a x) = 0.960253$ $n_a = 57, a = \{10, 9\}$: $C(x a) = 0.916370, C(a x) = 0.960253$ $n_a = 58, a = \{10, 10\}$: $C(x a) = 0.901830, C(a x) = 0.960253$
63	$\{11, 5\}$	$n_a = 56, a = \{10, 8\}$: $C(x a) = 0.956636, C(a x) = 0.960253$ $n_a = 57, a = \{10, 9\}$: $C(x a) = 0.940214, C(a x) = 0.960253$ $n_a = 58, a = \{10, 10\}$: $C(x a) = 0.924929, C(a x) = 0.960253$
64	$\{11, 6\}$	$n_a = 58, a = \{10, 10\}$: $C(x a) = 0.945667, C(a x) = 0.960253$

B.1 The global comparison clusters

On the infinite family of non-special integer signatures, we distinguish a non-strict and a strict comparison arrow:

$$a \longrightarrow b \iff C(a|b) \geq C(b|a), \quad a \xrightarrow{s} b \iff C(a|b) > C(b|a).$$

An arrow points from the greater element to the smaller one; equivalently, $a \xrightarrow{s} b$ means $b \prec a$. Equality produces two opposite non-strict arrows and no strict arrow. Define the corresponding clusters as the strongly connected components

$$\text{Cl}_{\geq}(a) = \{b : a \longrightarrow^* b \text{ and } b \longrightarrow^* a\},$$

and define $\text{Cl}_{>}(a)$ analogously using only $\xrightarrow{s}$. Reversing every arrow leaves these components unchanged.

Finite truncations require care. A path between two vertices of $\mathcal{S}_B$ may leave the shell, so a component of the induced subgraph need not be the intersection of a global component with $\mathcal{S}_B$. The 69-term prefix therefore gives components that are too small.

Calculations on nested shells suggest the following conjecture:

$$\text{Cl}_{\geq}(\{3, 1, 1\}) = \{a : |a| \geq 2, a_1 > 1\} \setminus \{\{2, 1\}, \{2, 2\}\}.$$

The computed component contains 112 of the 129 signatures in $\mathcal{S}_{12}$ and 2498 of the 2560 signatures in $\mathcal{S}_{18}$. It does not stabilize over $12 \leq B \leq 18$: enlarging the shell supplies new connecting paths. These counts support the conjecture but do not give a finite description of the global component.

For the strict graph, put

$$\begin{aligned} \mathcal{E}_s = \{ & \{2, 1\}, \{2, 2\}, \{3, 1\}, \{3, 2\}, \{3, 3\}, \{4, 1\}, \{4, 2\}, \{4, 3\}, \\ & \{2, 1, 1\}, \{2, 2, 1\}, \{2, 2, 2\}, \{2, 1, 1, 1\} \}. \end{aligned}$$

We conjecture that

$$\text{Cl}_>(\{3, 1, 1\}) = \{a : |a| \geq 2, a_1 > 1\} \setminus \mathcal{E}_s.$$

In $\mathcal{S}_{20}$, the computed strict component contains 6719 of 6738 signatures. Its intersection with $\mathcal{S}_{19}$ is $\mathcal{S}_{19} \setminus \mathcal{E}_s$; the other seven omitted vertices lie on the new boundary $a_1 + 2|a| = 20$. The computation does not prove that later shells absorb every boundary vertex, or that no element of $\mathcal{E}_s$ is eventually absorbed.

For reference, the following cycle was computed in $\mathcal{S}_{14}$ with tolerance $2 \cdot 10^{-6}$. A shortest strict path from $\{9, 1\}$ to $\{3, 2, 2, 2\}$ is

$$\{9, 1\} \xrightarrow{s} \{3, 3, 3\} \xrightarrow{s} \{3, 1, 1, 1\} \xrightarrow{s} \{8, 4\} \xrightarrow{s} \{10, 1\} \xrightarrow{s} \{3, 2, 2, 2\}.$$

Table 4 lists its edge rates.

Table 4: Strict path from $\{9, 1\}$ to $\{3, 2, 2, 2\}$.		
Strict edge $a \xrightarrow{s} b$	$C(a \mid b)$	$C(b \mid a)$
$\{9, 1\} \xrightarrow{s} \{3, 3, 3\}$	0.630929754	0.500000000
$\{3, 3, 3\} \xrightarrow{s} \{3, 1, 1, 1\}$	0.792481250	0.753940561
$\{3, 1, 1, 1\} \xrightarrow{s} \{8, 4\}$	0.527921767	0.500000000
$\{8, 4\} \xrightarrow{s} \{10, 1\}$	0.903089987	0.899142666
$\{10, 1\} \xrightarrow{s} \{3, 2, 2, 2\}$	0.500000000	0.477121255

A shortest return path has two edges:

$$\{3, 2, 2, 2\} \xrightarrow{s} \{7, 5\} \xrightarrow{s} \{9, 1\}.$$

Table 5: Strict return path from $\{3, 2, 2, 2\}$ to $\{9, 1\}$.

Strict edge $a \xrightarrow{s} b$	$C(a \mid b)$	$C(b \mid a)$
$\{3, 2, 2, 2\} \xrightarrow{s} \{7, 5\}$	0.564574819	0.500000000
$\{7, 5\} \xrightarrow{s} \{9, 1\}$	0.885621875	0.871797909

B.2 A length-three strict cycle

Strict cycles also occur at fixed length. Among the 55 non-special length-three signatures with entries at most 6, exhaustive computation at tolerance $2 \cdot 10^{-6}$ finds none. Allowing entries up to 7 gives

$$\{6, 3, 3\} \xrightarrow{s} \{7, 2, 1\} \xrightarrow{s} \{6, 5, 1\} \xrightarrow{s} \{6, 3, 3\}.$$

Table 6 gives both directed rates on every edge.

Table 6: A strict three-cycle of length-three signatures.

Strict edge $a \xrightarrow{s} b$	$C(a \mid b)$	$C(b \mid a)$	difference
$\{6, 3, 3\} \xrightarrow{s} \{7, 2, 1\}$	0.920782221	0.910910734	0.009871488
$\{7, 2, 1\} \xrightarrow{s} \{6, 5, 1\}$	0.924704131	0.920782221	0.003921910
$\{6, 5, 1\} \xrightarrow{s} \{6, 3, 3\}$	0.975533650	0.959474358	0.016059292